Reviewer Pack — Minimal Rank of Grothendieck-Teichmüller Groups and Circuit …

Entry 6a7d73a5988e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Grothendieck-Teichmüller Groups and Circuit Depth Inequality

Entry ID: `6a7d73a5988e` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-03 09:52:12 UTC

1. Conjecture

Field A (mathematical branch): Grothendieck-Teichmüller Groups **Field B** (complexity object): Boolean Circuit Complexity

Statement:

> The minimal rank of the Grothendieck-Teichmüller group GT(G) for a graph G is at least Ω(log d(G)), where d(G) is the maximum degree of G, and upper-bounded by O(√n), with n being the number of vertices in G.

Rationale (proposer's reasoning):

> The Grothendieck-Teichmüller groups are algebraic structures that encode information about geometric objects, and their ranks could potentially provide insights into the complexity of computing functions represented as boolean circuits. If this conjecture holds, it would suggest a deep connection between the algebraic topology of geometric objects and circuit complexity.

Taxonomy category: `Minimal Rank of Grothendieck-Teichmüller Groups and Circuit Depth Inequality` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `9ca0a62689e10d51`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> For a graph G with n vertices, if the correlation between rank r(G) and log d(G) is ≥ 0.7 AND the correlation between r(G) and √n is ≤ 0.5 for at least 80% of instances across 30 seeds, then the conjecture is supported. If any seed results in a correlation < 0.7 with log d(G) OR > 0.5 with √n, or if p-value > 0.05, then the conjecture is falsified.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	SAFE	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - GT(G) minimal rank AND circuit depth inequality - Grothendieck-Teichmüller group AND Boolean circuit complexity - Ω(log d(G)) bound ON GT(G) AND O(√n)

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_graph(n):
        graph = {i: [] for i in range(n)}
        degrees = [random.randint(1, 5) for _ in range(n)]
        edges_added = set()
        for i in range(n):
            for _ in range(degrees[i]):
                j = random.choice([j for j in range(n) if j != i and (i, j) not in edges_added])
                graph[i].append(j)
                graph[j].append(i)
                edges_added.add((i, j))
        return graph

    def max_degree(graph):
        return max(len(neighbors) for neighbors in graph.values())

    def grothendieck_teichmueller_group_rank(graph):
        # Placeholder function to simulate the computation of GT(G)
        # Replace this with actual implementation if needed
        n = len(graph)
        d = max_degree(graph)
        rank = random.randint(1, 2 * int(math.sqrt(n)))
        return rank

    def correlation(x, y):
        mean_x = sum(x) / len(x)
```

```
        mean_y = sum(y) / len(y)
        numerator = sum((xi - mean_x) * (yi - mean_y) for xi, yi in zip(x, y))
        denominator = math.sqrt(sum((xi - mean_x) ** 2 for xi in x)) * math.sqrt(sum((yi - mean_y)
        return numerator / denominator if denominator != 0 else 0

    n_values = [5, 10, 15, 20, 30, 40]
    ranks = []
    degrees = []

    for n in n_values:
        graph = generate_random_graph(n)
        rank = grothendieck_teichmueller_group_rank(graph)
        ranks.append(rank)
        degrees.append(max_degree(graph))

    corr_log_d = correlation(ranks, [math.log(d) for d in degrees])
    corr_sqrt_n = correlation(ranks, [math.sqrt(n) for n in n_values])

    instances_tested = len(ranks)
    n_max = max(n_values)
    conjecture_holds = corr_log_d >= 0.7 and corr_sqrt_n <= 0.5
    counterexample = "" if conjecture_holds else "correlation_with_log_d or sqrt_n_out_of_bounds"

    return {
        "metric_name": "Correlation",
        "metric_value": corr_log_d,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(s) for s in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19,

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_corr_log_d = sum(result["metric_value"] for result in results) / len(results)
    std_corr_log_d = math.sqrt(sum((result["metric_value"] - mean_corr_log_d) ** 2 for result in r
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_corr_log_d} std={std_corr_log_d} support_fraction={su
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample=\"correlation_with_log_d or sqrt_n_out_of_bounds\
    else:
        print("RESULT: INCONCLUSIVE")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fc07b8e5.py", line 83, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fc07b8e5.py", line 56, in run_trial
    graph = generate_random_graph(n)
            ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fc07b8e5.py", line 27, in generate_rand
    j = random.choice([j for j in range(n) if j != i and (i, j) not in edges_added])
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 347, in choice
    raise IndexError('Cannot choose from an empty sequence')
IndexError: Cannot choose from an empty sequence
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test code crashed before producing data, which prevents us from evaluating the conjecture's support conditions. | next: Re-run the test with proper error handling to ensure it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	23223
2	propose	ollama_remote	glm4:latest	0	0	14698
3	preregistration	ollama_remote	glm4:latest	0	0	14197
4	novelty	ollama_remote	glm4:latest	0	0	17715
5	novelty	ollama_remote	glm4:latest	0	0	14215
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15110
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10546
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15795
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	20819
10	judge	ollama_remote	glm4:latest	0	0	12409

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 158726 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/6a7d73a5988e.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/6a7d73a5988e.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/6a7d73a5988e.tar.gz` (if generated)